Cloud detection on Sentinel-2 images

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1. Intro and rationale

Clouds and various kinds of cloud cover are a major obstacle in Earth Observation and optical remote sensing in particular. Global observations from the Moderate Resolution Imaging Spectroradiometer (MODIS) indicate an average cloud fraction of approximately 67%, with cloud cover averaging about 55% over land and 72% over oceans with the distinct seasonal cycle (King et al., 2013). Different types of clouds appear differently in the satellite imagery. For example thick clouds (e.g., cumulonimbus clouds) typically appear bright and opaque and may be confused with snow, ice, or other bright surfaces, whereas thin high altitude clouds (e.g., cirrus) can be semi-transparent and may be difficult to distinguish due to the weaker contrast with the surface (Li et al., 2022).

Algorithms of various complexity have been used to identify cloud masks on optical images. Commonly used operational cloud masking algorithms rely on both rule-based and deep-learning approaches, depending on the sensor. For Sentinel-2 Level-2A imagery, Sen2Cor algorithm generates a scene classification layer using spectral threshold tests and spatial and terrain-based processing to identify clouds, thin cirrus, cloud shadows, and surface classes (ESA, 2021). PlanetScope's Usable Data Mask version 2.1 (UDM2.1) uses a modified U-Net segmentation model applied to four-band RGB–near-infrared imagery to distinguish clear pixels, clouds, haze, cloud shadows, and snow (Planet).

In this report I present a machine learning pixel-based framework to automate identification of clouds on Sentinel-2 optical satellite imagery and explore 2 different strategies for sampling that dataset for the training data.

2. Brief data description

I use the labelled dataset derived from Sentinel-2 images provided by (Hollstein et al., 2016), version 2017-07-10. It contains 3,028,733 labelled pixels[1], each described by 13 spectral bands from 443 to 2190 nm, with no missing or negative values. The spectra are ordered B1–B8, B8A, B9–B12, so B8A is the ninth component.

Classes. Each pixel belongs to one of six classes (Figure 1B). The dataset contains unequal number of pixels per class, therefore class frequencies reflect the labelling rather than how often each class occurs in real imagery. Reported precision and probabilities should be interpreted with that in mind.

Scenes. The pixels come from 48 scenes and 51 granules (100x100 km image tiles) (Figure 1A). Some granules appear in more than one product, so I merge linked products and granules into 39 scene groups. Group is the effective unit of independence: pixels within a scene share atmospheric and illumination conditions, so the effective sample size is much smaller than the pixel count. Products also vary widely in size, from 4,345 to 218,694 pixels.

[1] The assessment brief states about 5.6 million pixels, while the 2017-07-10 file used here contains 3,028,733.

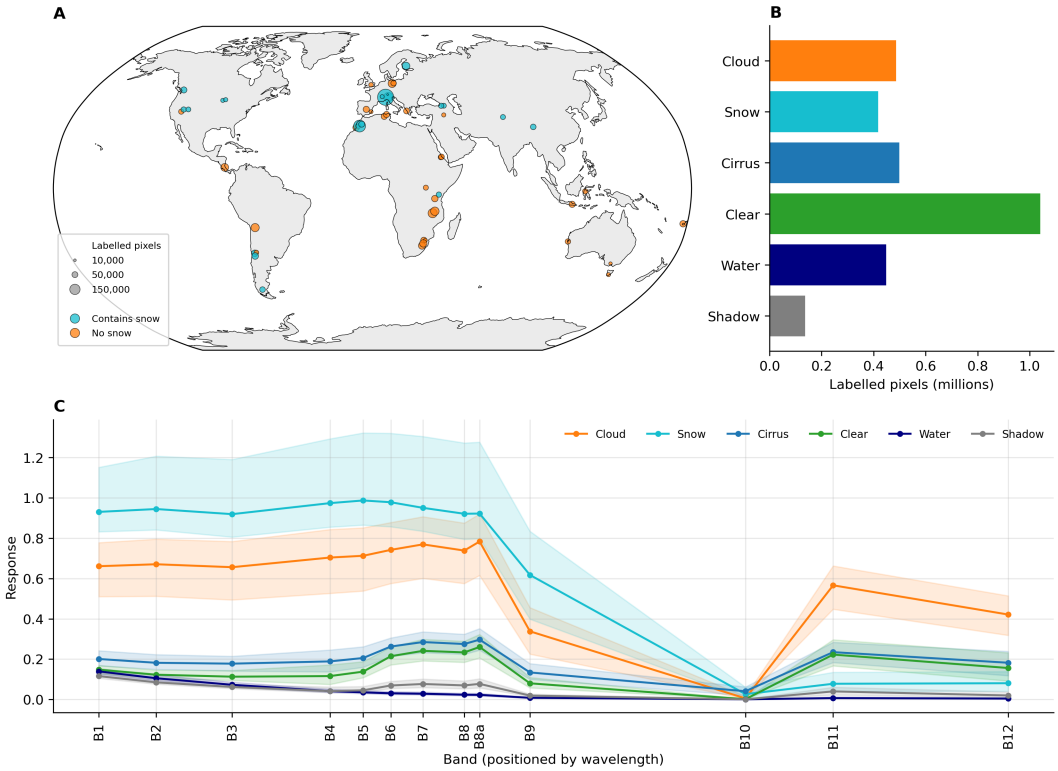


Figure 1: Overview of the data. (A) Geographical distribution of labelled pixels, (B) Number of pixels per class in the training dataset, (C) Spectral signatures of classes in the dataset.

Spectral behavior. Figure 1C shows that the classes have distinct spectral signatures. Cloud and snow are both bright in the visible bands, but snow is much darker in SWIR (median B11 of 0.08, compared with 0.57 for cloud). B10 has a very low median across all pixels (0.002), consistent with strong atmospheric absorption, and it helps to distinguish cirrus from clear pixels. We therefore treat B10, rather than B11 as described in the brief, as the absorbed cirrus band. Responses above 1 in bands B1–B8A occur only in the cloud (15%) and snow (49%) classes, consistent with very bright targets, so we retained them.

3. Modelling experiments

I set up a modelling experiment as a per-pixel classification problem, using the 13 spectral bands as inputs and four target classes: Cloud, Snow, Water and Rest (clear, shadow and cirrus). Snow was kept separate because I expect it to be confused with the cloud due to high brightness in visible bands (however, they have different response in the shortwave infrared) (Stillinger et al., 2019). Water was kept separate due to commonly being confused with snow (Hall et al., 1998). Cirrus was assigned to Rest, so the Cloud class represents opaque cloud. I use CatBoost model (Dorogush et al., 2018; Prokhorenkova et al., 2018) - a simple gradient boosted model and build a classification framework using the target classes as an output. I evaluate the model under two sampling strategies to measure how much the validation design affects the performance.

3.1. Validation strategy

Pixels from the same scene share atmospheric and illumination conditions, and because labels were drawn as polygons, many pixels are located in close proximity to each other. Therefore, treating them as independent samples may overestimate performance on new imagery and lead to overfitting. I linked data coming from products that share a granule into 39 scene groups, and used these as the unit of independence (Figure 1).

Before the modelling experiments, I set aside 9 scene groups (575,439 pixels) as a holdout, selected using class coverage only. The remaining 30 groups (2,453,294 pixels) formed the training and testing set, on which both sampling strategies were compared using 5-fold cross-validation:

- **Random pixel sampling:** pixels were assigned to folds at random, stratified by class, ignoring scene membership. Pixels from one scene could therefore appear in both training and test data.
- **Scene-group sampling:** folds were built with stratified group 5-fold splitting, keeping each scene group entirely within training, validation or test data.

In each fold, one-fifth of the non-test data, selected by pixel or by scene group according to the strategy, formed a validation set for CatBoost early stopping (up to 500 trees, depth 6, learning rate 0.1). Training and validation data were limited to 30,000 pixels per scene group to limit the over-influence of large scenes, while test folds remained complete. Evaluation metrics are calculated using all 5 folds. Random sampling gave models access to pixels from more scenes during training (all 30 groups, compared with 16-25), so differences between the strategies reflect training coverage as well as data leakage. The holdout was used once, after the model was built and trained, to give an independent estimate of performance on the completely unseen data. The classification threshold is set to 0.5.

4. Results

Table 1: Cloud detection at $P(\text{Cloud}) \geq 0.5$ under random and scene-based sampling. Parentheses show 95% intervals from resampling whole scene groups. Cirrus is counted as not cloud. TP, FN, FP and TN are true positives, false negatives, false positives and true negatives.

	Random		Scene	
Metric	CV	Holdout	CV	Holdout
Recall	0.994 (0.99–1.00)	0.941 (0.72–0.99)	0.920 (0.86–0.97)	0.930 (0.70–0.99)
Precision	0.996 (0.99–1.00)	0.996 (0.98–1.00)	0.910 (0.79–0.97)	0.998 (0.99–1.00)
PR-AUC	>0.999	0.995 (0.94–1.00)	0.981 (0.96–0.99)	0.995 (0.95–1.00)
TP	391,964	86,552	362,767	85,569
FN	2,445	5,469	31,642	6,452
FP	1,383	352	35,700	140
TN	2,057,502	483,066	2,023,185	483,278

The model performance is presented in Table 1 and Figure 2. Sampling strategy strongly affected the cross-validation estimates but had little effect on holdout performance (Table 1). Under random sampling strategy, cloud

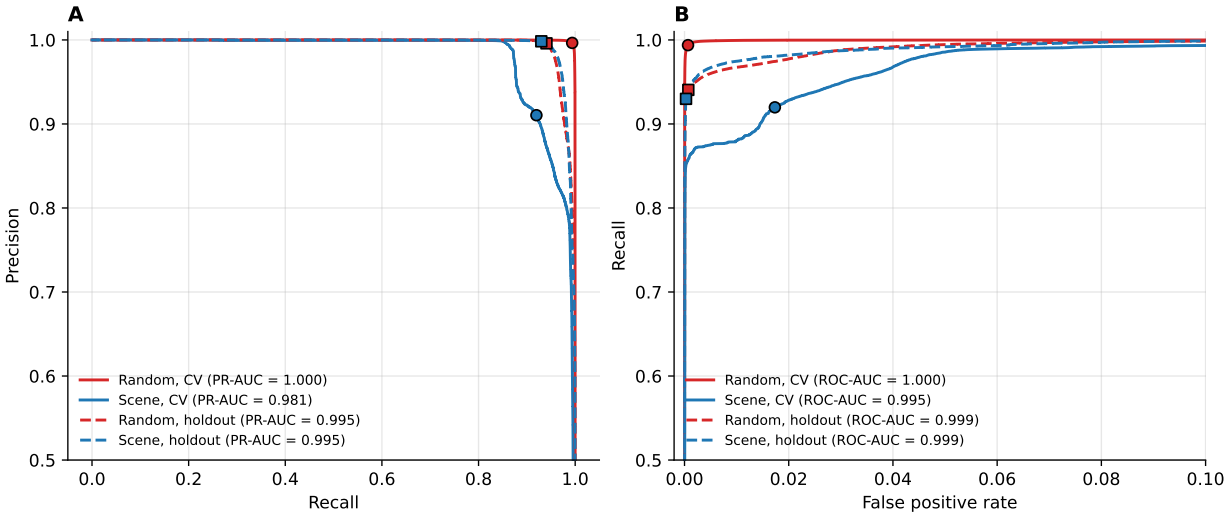


Figure 2: Precision and Recall and ROC AUC curves for different sampling strategies and the hold out set.

recall was 0.994 in cross-validation but 0.941 on the holdout, indicating that performance on new scenes might be overestimated. Scene-based cross-validation gave a recall of 0.920, which is closer to holdout recall of 0.930. Both final models performed similarly on the holdout, with overlapping intervals for recall (0.72–0.99 and 0.70–0.99), precision (0.98–1.00 and 0.99–1.00) and PR-AUC (0.94–1.00 and 0.95–1.00). However, scene-based cross-validation did not match the holdout on every metric: its precision of 0.910 was below the holdout value of 0.998, meaning fewer false cloud detections among cirrus pixels in the holdout scenes. Scene-based sampling strategy does not produce a better performing model, but it provides a more cautious estimate of performance on unseen scenes.

At $P(\text{Cloud}) \geq 0.5$, cirrus accounted for most false positive cloud detections in scene-based cross-validation (65%), followed by clear pixels (25%). Snow, accounted for 6% (despite my hypothesis stating snow and cloud similarity), and water for 4%. In the holdout, there were only 140 false detection instances, including 35 from cirrus.

I applied the model trained with the scene-based strategy to a Sentinel-2 Level-1C image (top-of-atmosphere reflectance), matching the processing level of the training data. The image is tile T45RTM, 5 November 2026, resampled to 60 m (Figure 3), Central Himalaya. Clouds as well as snow are present in the image. The model assigned $P(\text{Cloud}) \geq 0.5$ to 7.6% of pixels, and 3.7% had intermediate values between 0.2 and 0.8. Visually, the cloud mapping using the trained CatBoost model works really well, accurately identifying snow cover and clouds (Figure 3). However, this particular image does not have labelled data to quantify the model performance, therefore only visual inspection is possible.

5. Discussion

Even though CatBoost model achieves a very high level of performance, several limitations affect how these results may be generalized. Pixels in the training data were manually selected, therefore ambiguous cases such as

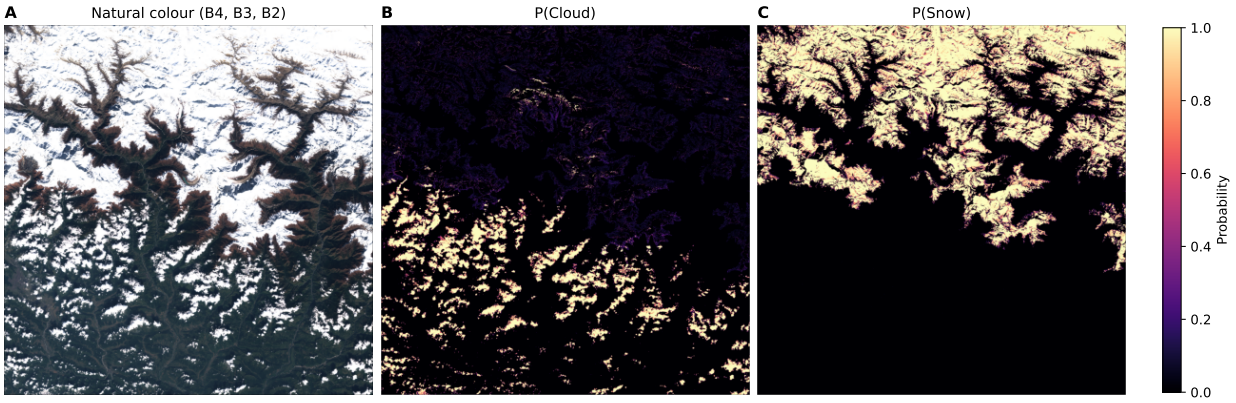


Figure 3: Evaluation of the model on the real Sentinel-2 image not included in the training and hold out set (Central Himalaya, tile T45RTM, 5 November 2026).

cloud edges and thin cloud may be underrepresented, and class frequencies do not reflect real imagery. Scores may therefore overestimate performance and the model should be evaluated on a larger spatially diverse areas. Cirrus pixels were defined as not-cloud. It can still contaminate surface reflectance. Much of the false detection in scene-based cross-validation came from pixels that have a cirrus label, so results depend on the definition of a cloud. As well as this, the model classifies each pixel independently, without using spatial context.

Code availability

Code is available at https://github.com/varyabazilova/cloud_detection.

AI assistance

I used Claude Opus 5.5 (Anthropic) throughout this assessment to discuss the validation design, draft and debug analysis code. The ideas, hypothesis and interpretations are entirely my own. I reviewed the code, interpreted results and wrote the text, and take responsibility for the final report.

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